

Technical Report: Complete Business Analysis Project

This technical report presents a comprehensive end-to-end business analysis using three real-world datasets: house prices, customer churn, and sales data. The objective is to extract meaningful insights through data analysis and visualization, and translate them into actionable business recommendations.

Dataset Description

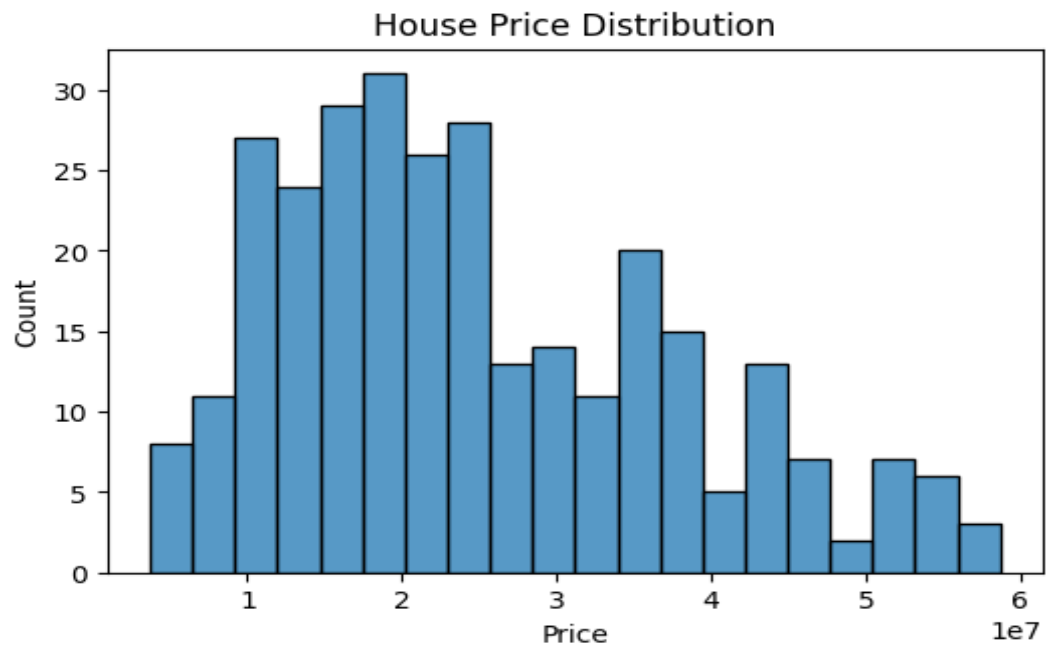
1. House Prices Dataset: Contains information on property area, number of bedrooms, bathrooms, age, and price.
2. Customer Churn Dataset: Includes customer churn status and monthly charges.
3. Sales Dataset: Records transactional sales data across regions and time.

Data Cleaning Process

Data cleaning involved handling missing values, standardizing column names, converting data types, and removing inconsistencies. Only numeric columns were used for statistical analysis such as correlation.

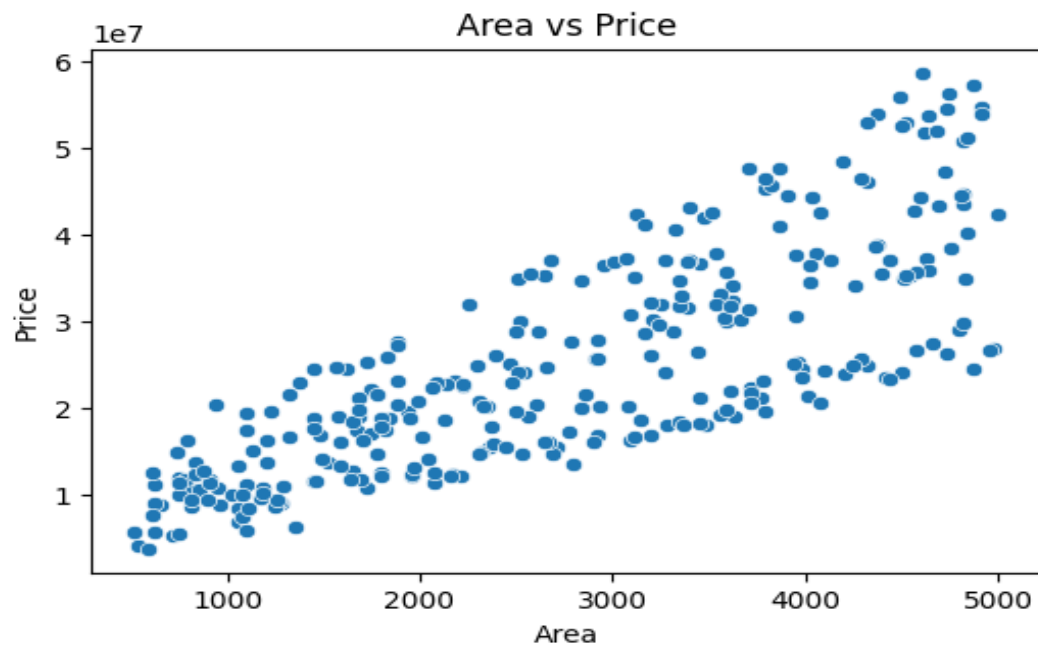
Exploratory Data Analysis – House Prices

The house price distribution shows a right-skewed pattern, indicating a higher frequency of mid-range properties with fewer high-end luxury properties.



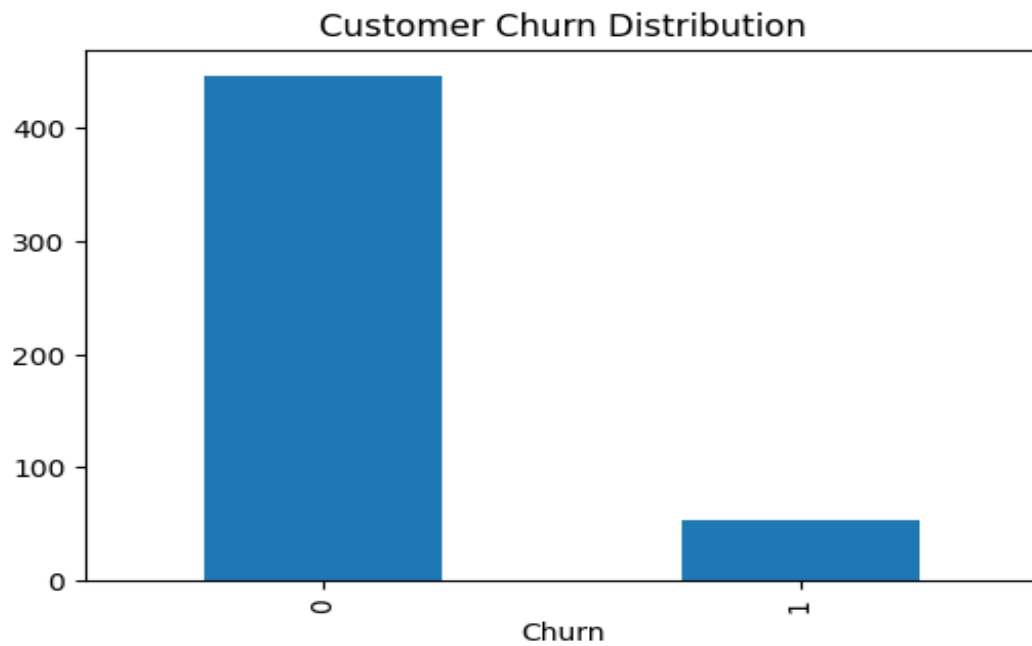
Area vs Price Analysis

A strong positive relationship is observed between property area and price, suggesting that area is a major price-determining factor in the real estate market.



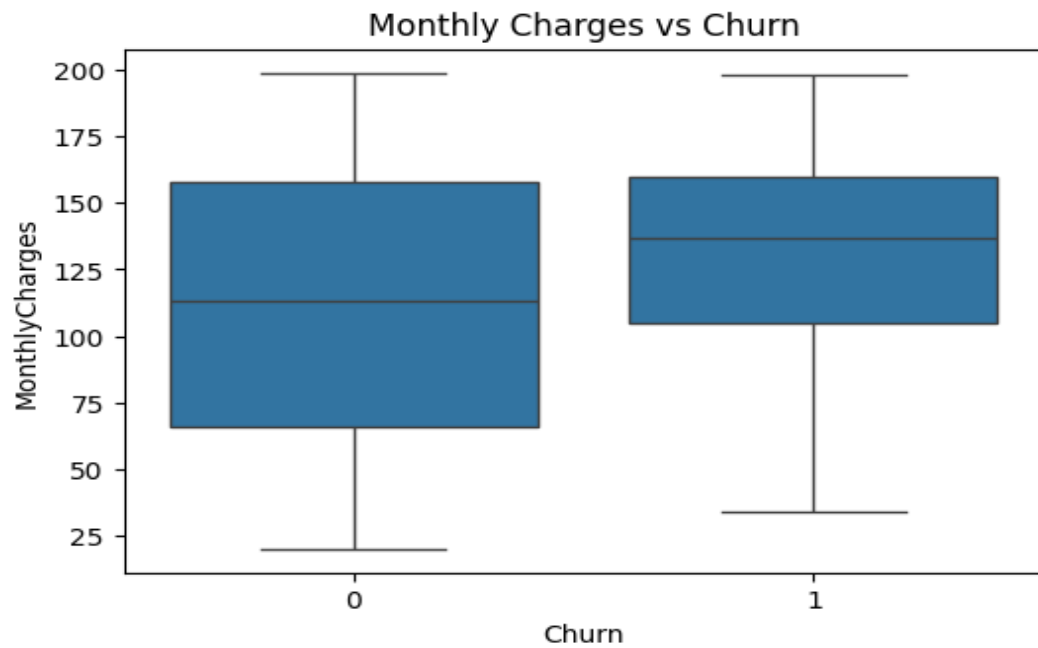
Customer Churn Analysis

The churn distribution indicates that a smaller proportion of customers churn compared to those who remain. However, churned customers contribute significantly to revenue loss.



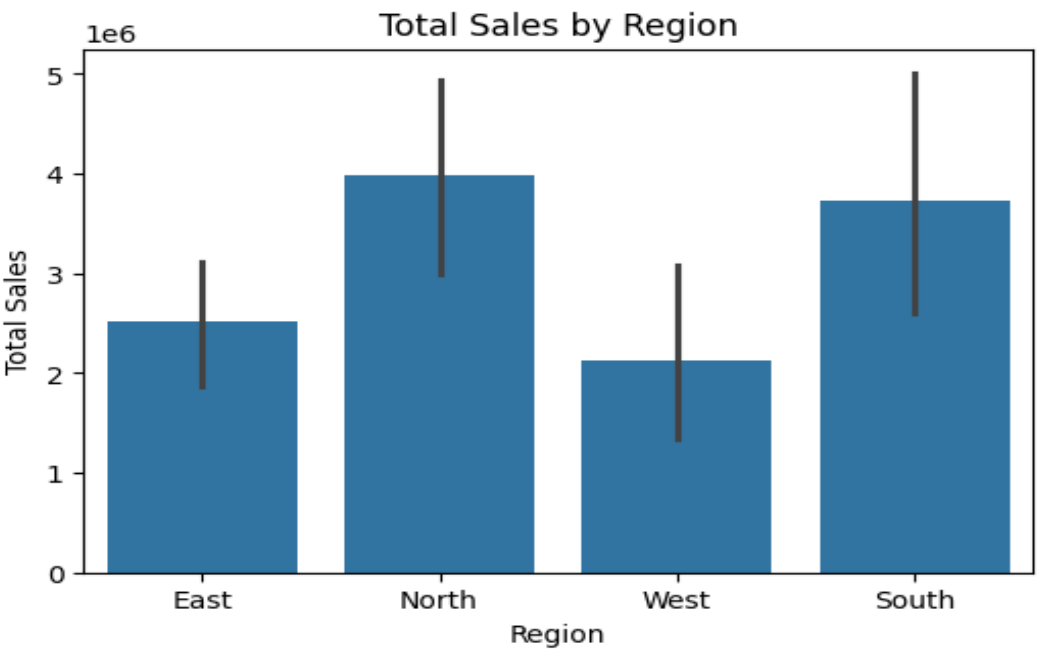
Monthly Charges vs Churn

Customers with higher monthly charges are more likely to churn, highlighting pricing sensitivity and the need for targeted retention strategies.



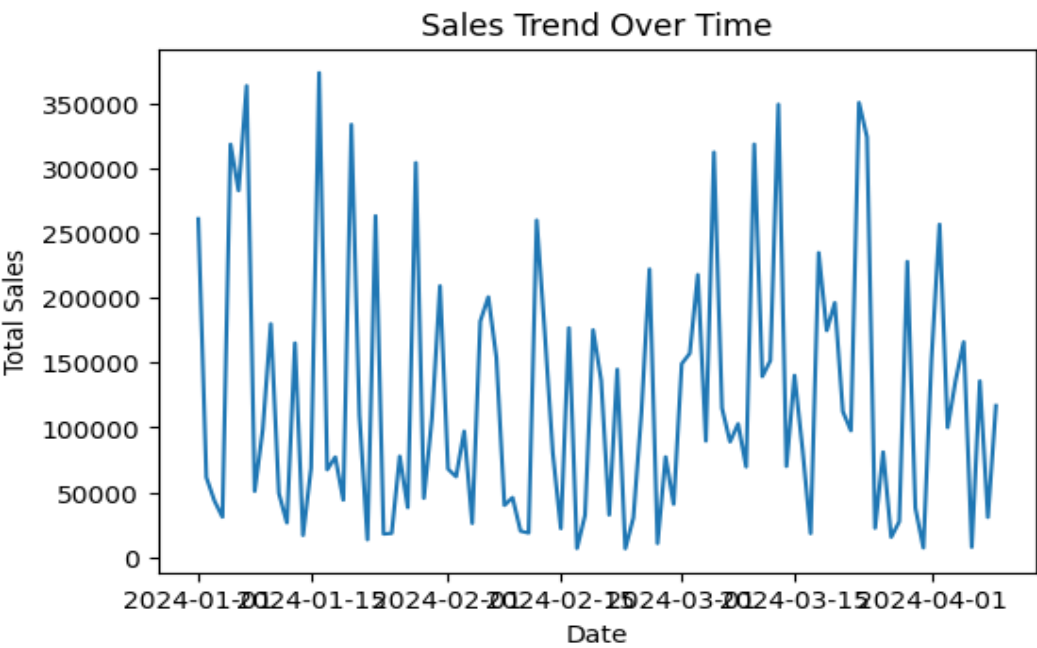
Sales Performance by Region

Sales analysis reveals that certain regions consistently outperform others, indicating regional differences in demand and market potential.



Sales Trend Over Time

The sales trend over time shows fluctuations with periodic spikes, suggesting the influence of seasonal factors, promotions, or demand cycles.



Advanced Analysis & Conclusion

Correlation analysis indicates that house price has a strong positive correlation with area, while other features show moderate to weak relationships. Overall, the project demonstrates how data-driven analysis can support strategic decision-making across multiple business domains.

